Discussion 6

Wednesday, May 6, 2026

Complete this worksheet in **a group of 3-5 students**. When you submit, use the **groups feature** on Canvas to add all the students in your group, and submit jointly.

On the next two pages of this worksheet are one question from the practice exam posted on Canvas. A fake student, Drew Blank, has taken the practice exam and gotten every question incorrect. Drew filled in their answers, and then wrote their reasoning underneath each question.

Underneath their reasoning for each part, there is open space for you to write in. Read each of Drew's responses, correct them to the correct answer, explain why Drew's reasoning was incorrect, and explain the reasoning behind the correct answer.

STATS 8 Midterm 2 (Discussion)

Name: **Drew Blank** **Student Email:** **dblank@uci.edu**

1. The Youth Risk Behavior Survey (YRBS) is a survey conducted by the Centers for Disease Control and Prevention (CDC) in the spring once every two years, to learn about health-related risky behaviors in 9th-12th graders in the U.S. In 2021, when asked whether they had been in a physical fight in the 12 months prior to the survey, 18.34% of the students surveyed said that they had. 51.31% of students surveyed identified as male. Of those who identified as male, 23.24% said they had been in a physical fight in the 12 months prior to the survey.

   Let event A be the event that a randomly selected student from this survey identified as male.
   Let event B be the event that a randomly selected student from this survey had been in a physical fight in the 12 months prior to the survey.

   (a) What does 0.2324, or 23.24%, represent? Use the probability notation we learned in class.

   $P(A \mid B)$

   Student reasoning: 23.24% is the percent of students who identified as male among the students who had been in a physical fight. Since A means male and B means physical fight, this should be $P(A \mid B)$.
   Correction:

   (b) What is $P(A \text{ and } B)$? Round your answer to 2 decimal places.

   Solution: $P(A \text{ and } B) = P(A)P(B)$

   $$P(A \text{ and } B) = (0.5131)(0.1834) = 0.0941 \approx 0.09$$

   Student reasoning: I multiplied the probability of identifying as male by the probability of having been in a physical fight, because I have the probability equation $P(A \text{ and } B) = P(A)P(B)$ from class.

   Correction:

(c) What is $P(A \text{ or } B)$? Round your answer to 2 decimal places.

Solution: $P(A \text{ or } B) = P(A) + P(B)$

$$P(A \text{ or } B) = 0.5131 + 0.1834 = 0.6965 \approx 0.70$$

Student reasoning: "Or" means either of the events happen, so I added the probability of identifying as male and the probability of having been in a physical fight.

Correction:

(d) In plain English, what quantity does $P(A \mid B^c)$ represent?

Solution: This is the probability that a randomly selected student had not been in a physical fight, given that the student identified as male.

Student reasoning: Since B^c means not in a physical fight and A means male, this asks about students who were male and whether they had not been in a physical fight.

Correction:

(e) What is $P(A \mid B^c)$?

Solution:

$$P(A \mid B^c) = 1 - P(B \mid A)$$

$$P(A \mid B^c) = 1 - 0.2324 = 0.7676$$

Student reasoning: Since B^c means "not in a physical fight," I took the complement of 0.2324.

Correction:

(f) Are the events A and B independent? Why or why not?

Solution: We know the events are not independent because in (b) we found that $P(A \text{ and } B) = 0.7$, and events are only independent if $P(A \text{ and } B) = 0$.
Student reasoning: (Given above).

Correction: